

Assignment No 8

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Aim: (Functions)

Write a program to build a simple Employee and Project management System using Python functions which can perform the following operations:

1. Accept [Project name, Team members, Technologies used, Project completion deadline]
2. Display - displays the details of every Project
3. Search - Look for a specific employee and the project on which he or she is working
4. Delete - delete a particular record (based on either project/employee)
5. Update - Update the team members for a particular project

Theory:

To create a connection between the MySQL database and Python, the connect() method of mysql.connector module is used. We pass the database details like HostName, username, and the password in the method call, and then the method returns the connection object.

The following steps are required to connect SQL with Python:

Step 1: Download and Install the free MySQL database

Step 2: After installing the MySQL database, open your Command prompt.

Step 3: Navigate your Command prompt to the location of PIP

Step 4: Now run the commands given below to download and install “MySQL Connector”. Here, mysql.connector statement will help you to communicate with the MySQL database.

Download and install “MySQL Connector”

```
pip install mysql-connector-python
```

Step 5: Test MySQL Connector

To check if the installation was successful, or if you already installed “MySQL Connector”, go to your IDE and run the given below code :

```
import mysql.connector
```

Step 6: Create Connection

Now to connect SQL with Python, run the code given below in your IDE.

```
# Importing module
```

```
import mysql.connector
```

```
# Creating connection object
```

```
mydb = mysql.connector.connect(
```

```
    host = "localhost",
```

```
    user = "yourusername",
```

```
    password = "your_password"
```

```
)
```

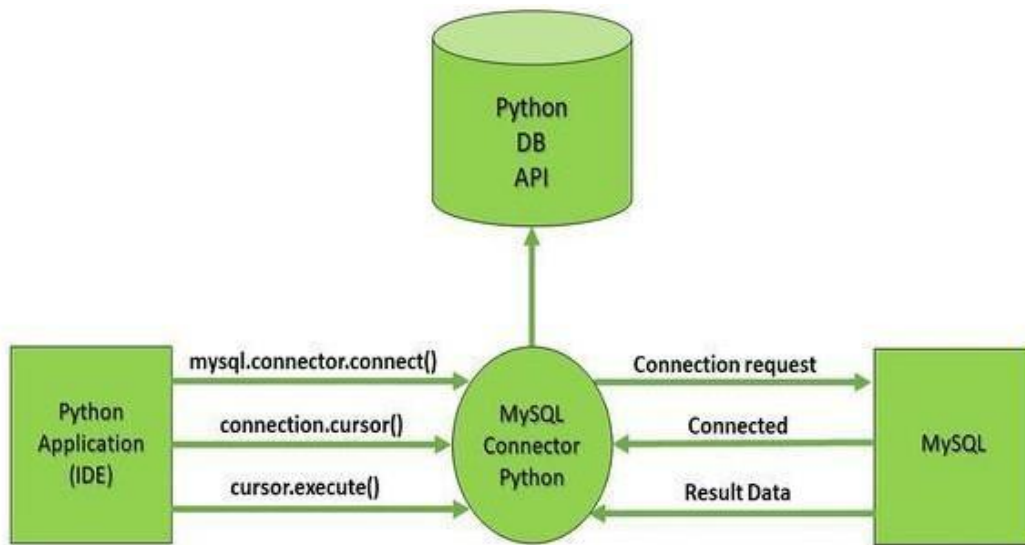
```
# Printing the connection object
```

```
print(mydb)
```

Creating MySQL Database

To create a database, we will use `CREATE DATABASE database_name` statement and we will execute this statement by creating an instance of the ‘cursor’ class.

Now to create tables in a database, first, we have to select a database and for that, we will pass `database = “NameofDatabase”` as your fourth parameter in `connect()` function.



Program:

```
import pymysql

# pip3 install pymysql

def get_connection():

    connection = pymysql.connect(host = 'localhost', user='root', password='Admin@123',
database='project_management')

    return connection

def accept(id, name, team_members,technology, deadline):

    connection = get_connection()
    cursor = connection.cursor()

    query = "insert into projectinfo values ({},{},\{'\','\','\','\','\','\'});".format(id, name,
team_members, technology,deadline)

    cursor.execute(query)
    connection.commit()

    print("Record Added Successfully")

accept(2, "Video proctoring", "Abhijeet", "C#", "March 2023")

def display():

    connection = get_connection()
    cursor = connection.cursor()

    query = "select * from projectinfo;"

    cursor.execute(query)

    data = cursor.fetchall()
```

```
    for i in data:
        print(i)
display()
```

```
def search(project_name):
    connection = get_connection()
    cursor = connection.cursor()
    query = "select * from projectinfo where name = '{}'\".format(project_name)
    cursor.execute(query)
    data = cursor.fetchone()
    print(data)
print("search results:")
search("Video proctoring")
```

```
def update(id, name, team_members, technology, deadline):

    connection = get_connection()
    cursor = connection.cursor()
    query = "update projectinfo set name = '{}', members= '{}', technologies_used = '{}',
    deadline = '{}'\ where projectid = {}".format(name, team_members, technology,
    deadline, id)
    cursor.execute(query)
    connection.commit()

    update(2, "Video proctoring", "Smita", "React Native", "DEC 2023")
print("After updating:")
display()
```

```
def delete(id):
```

```

connection = get_connection()
cursor = connection.cursor()
query = "delete from projectinfo where projectid = {}".format(id)
cursor.execute(query)
connection.commit()

```

```

delete(2)
print("After deleting :")
display()

```

Output: Output on mysql database:

```

mysql> select * from projectinfo;
+-----+-----+-----+-----+-----+
| projectid | name           | members | technologies_used | deadline |
+-----+-----+-----+-----+-----+
| 1 | Image Proctoring | Mandar  | Reactjs          | july 2025 |
+-----+-----+-----+-----+-----+
row in set (0.01 sec)

mysql> select * from projectinfo;
+-----+-----+-----+-----+-----+
| projectid | name           | members | technologies_used | deadline |
+-----+-----+-----+-----+-----+
| 1 | Image Proctoring | Mandar  | Reactjs          | july 2025 |
| 2 | Video Proctoring | Nilesh  | C#               | March 2023 |
+-----+-----+-----+-----+-----+
rows in set (0.00 sec)

mysql> select * from projectinfo;
+-----+-----+-----+-----+-----+
| projectid | name           | members | technologies_used | deadline |
+-----+-----+-----+-----+-----+
| 1 | Image Proctoring | Mandar  | Reactjs          | july 2025 |
| 2 | Video Proctoring | Smita   | React Native     | DEC 2023  |
+-----+-----+-----+-----+-----+
rows in set (0.00 sec)

mysql> select * from projectinfo;
+-----+-----+-----+-----+-----+
| projectid | name           | members | technologies_used | deadline |
+-----+-----+-----+-----+-----+
| 1 | Image Proctoring | Mandar  | Reactjs          | july 2025 |
+-----+-----+-----+-----+-----+

```

(1,'Image Proctoring', 'Mandar', 'React JS', 'July 2025')

(2, "Video proctoring", "Abhijeet", "C#", "March 2023")

Search results:

(2, "Video proctoring", "Abhijeet", "C#", "March 2023")

After updating:

(1,'Image Proctoring', 'Mandar', 'React JS', 'July 2025')

(2, "Video proctoring", "Smita", "React Native", "DEC 2023")

After deleting :

(1,'Image Proctoring', 'Mandar', 'React JS', 'July 2025')

Conclusion: We have successfully learnt how to create data base connections in python and perform various operations on it.